



**BAHIR DAR UNIVERSITY INSTITUTE OF
TECHNOLOGY
FACULTY OF COMPUTING
DEPARTMENT OF SOFTWARE ENGINEERING
Project Proposal**

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Final Year Project Idea #1 (Project Fidel: A Digital Gateway to Ethiopian Languages)

Introduction

Fidel is a streamlined, web-based platform designed to provide an accessible entry point into Ethiopian languages. For our initial phase, the platform offers a focused, interactive curriculum centered on the Amharic script (Fidel) and essential vocabulary. The platform will be built as a lightweight Progressive Web App using a modern JavaScript stack. By focusing on a clean User Interface (UI) and a robust database for lesson content, we are establishing a scalable foundation that proves Ethiopian languages can be taught effectively through modern web standards.

Statement of the Problem

Existing digital resources for Ethiopian language literacy are currently fragmented and outdated, primarily consisting of static PDFs and text-heavy websites that lack the interactivity required for modern learners. High entry barriers further complicate access, as many available mobile applications require significant storage space and high-speed data luxuries often unavailable in the local context. Moreover, the inherent complexity of the 200+ characters in the Fidel script remains a major hurdle for beginners, yet few platforms provide the structured, character-by-character digital practice necessary for mastery.

Objectives

- I. **Main Objective:** The primary goal of Project Fidel is to provide an interactive and accessible platform for learning the Ethiopian Fidel script and basic linguistic foundations.
- II. **Specific Objective:**
 - Create an interactive "Character-to-Phonetic" module that helps users master the vowel of the Ethiopic alphabet through visual and auditory cues.
 - Design a backend database structure that allows for the seamless addition of more Ethiopian regional languages

Significance of the Project

Significance of Fidel lies in its role as a digital bridge between Ethiopia's ancient linguistic heritage and the modern web. By developing a browser-based platform, this project addresses the critical need for accessible, high-quality indigenous language tools that do not require high-end hardware or expensive data plans. For the Ethiopian diaspora, it provides a vital link to their cultural identity, while for domestic users, it fosters national unity through cross-regional language appreciation. Ultimately, Fidel ensures that the unique phonetic and calligraphic beauty of Ethiopia's 80+ languages is not lost in the digital divide but is instead preserved and promoted for a new generation of learners.

Final Year Project Idea #2 (Private Git-Based Collaboration Platform for Academic Projects [PGCP])

Introduction

PGCP is a specialized, lightweight version of a Git-based version control system, reimagined specifically for the classroom. Our platform offers a secure, institutional environment where instructors can distribute assignments, students can collaborate in isolated teams, and the entire academic lifecycle 'from the first commit to the final grade' is managed in one place. Unlike generic platforms like GitHub or GitLab, which are often cluttered with complex DevOps tools, PGCP will focus on simplicity and pedagogical needs. By hosting the platform on private institutional servers, we tend to provide full data sovereignty, ensuring that student work remains private and under the school's control.

Statement of the Problem

Current Git platforms are often ill-suited for academic environments due to their focus on professional DevOps workflows, which introduce unnecessary complexity for students and high administrative overhead for instructors. In a classroom setting, public cloud-hosted repositories raise significant concerns regarding academic integrity and data privacy, as students may accidentally expose sensitive coursework. Furthermore, existing tools lack specialized "education-first" features such as automated assignment provisioning and submission locking at specific deadlines leaving a gap for a lightweight, secure, and private collaboration system designed specifically for the academic lifecycle.

Objectives

- **Primary Objective:** to develop a functional, private Git-based web platform that automates the distribution, collaboration, and submission of academic coding projects within a secure, role-based environment.
- **Specific Objectives:** To Implement Private Repository Hosting, To Develop Role-Based Access Control and To Enforce Submission Deadlines.

Significance of the Project

The PGCP platform transforms version control from a professional utility into a powerful pedagogical tool. For students, it provides a "sandbox" environment to learn the essentials of Git collaboration without the intimidation or public exposure of professional-grade platforms. For instructors, it eliminates the manual burden of managing dozens of repositories, allowing them to focus on code review and mentorship. For institutions, it ensures absolute data sovereignty and academic integrity by keeping all intellectual property on internal servers. Ultimately, this project bridges the gap between industry-standard practices and classroom needs, creating a streamlined ecosystem for modern technical education.

Final Year Project Idea #3 (Cultural Heritage Information Management System)

Introduction

CHIMS is a comprehensive, web-based platform designed to catalog, manage, and preserve Ethiopia's vast wealth of historical sites, artifacts, and intangible traditions. Our platform offers a centralized digital repository where high-resolution documentation, historical metadata, and geographical data are organized into an accessible "Digital Archive." It will be designed with a responsive frontend to ensure that researchers, curators, and the public can access this data from any device, anywhere in the world.

Statement of the Problem

Despite Ethiopia's status as a global heritage powerhouse, much of its cultural data remains vulnerable and inaccessible. Current information management suffers from a heavy reliance on paper-based records that are prone to physical decay and loss due to environmental factors. Furthermore, existing digital documentation is often fragmented across different regional offices, creating "data silos" that make it difficult for international researchers or domestic tourists to find accurate information. Without a centralized, cloud-secured platform, there is a persistent risk of losing irreplaceable historical data, and the barrier to accessing Ethiopia's cultural narrative remains unnecessarily high.

Objectives

- To develop a secure, centralized web-based Information Management System (IMS) that digitizes and streamlines the preservation, search, and retrieval of Ethiopian cultural heritage data.
- To Develop a role-based system where Admins manage sites, Researchers upload specialized documentation, and the Public can view curated history.
- To Enable Advanced Search and Filtering.
- To Implement automated cloud backup protocols to prevent the loss of historical records.

Significance of the Project

The development of this project is a vital step toward the "Digital Repatriation" and preservation of Ethiopian history. By digitizing heritage data, the project provides an insurance policy against the physical destruction of sites or the misplacing of paper records. For government agencies, it serves as a powerful tool for monitoring conservation efforts and managing tourism. For educators and researchers, it democratizes access to primary historical data that was previously locked in physical archives. Ultimately, this system ensures that Ethiopia's cultural legacy is not only preserved for future generations but is also shared with the global community, fostering a deeper understanding of the country's unique contributions to human history.